THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE FULL SYLLABUS PAPER SET 02

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The Moon appears to shrink after purnima because: 1

(a) a smaller lit part is visible from Earth **(b)** the Moon gets smaller
(c) the Sun goes out **(d)** Earth blocks all light

2. A prediction in science is: 1

(a) a testable expectation about what will happen
(b) a final law
(c) a proven fact
(d) a rumour

3. The main gas in biogas used as fuel is: 1

(a) chlorine **(b)** methane **(c)** oxygen **(d)** helium

4. Antibiotics are effective against: 1

(a) protozoan infections only **(b)** all diseases
(c) bacterial infections **(d)** viral infections

5. Ayurveda says true health is a balance of: 1

(a) only exercise **(b)** only diet
(c) body, mind and surroundings **(d)** only medicine

6. In a lemon cell, the electrolyte is: 1

(a) copper wire **(b)** lemon juice **(c)** iron nail **(d)** LED

7. Why should batteries not be thrown in ordinary garbage? 1

(a) they are too light **(b)** they contain harmful metals
(c) they are edible **(d)** they rot soon

8. An object sinks when gravitational force is: 1

(a) less than buoyant force **(b)** absent
(c) greater than buoyant force **(d)** equal to buoyant force

9. Mango showers occur in: 1

(a) Bihar only **(b)** Assam only
(c) Kerala, Karnataka and Tamil Nadu **(d)** Punjab

10. Faster wind speed means: 1

(a) zero air **(b)** no change
(c) higher air pressure **(d)** lower air pressure

11. Solids have a fixed: 1

(a) shape and volume **(b)** shape only **(c)** volume only **(d)** neither

12. Kamsya (bronze) had copper and tin in the ratio: 1

(a) 1 : 4 **(b)** 2 : 1 **(c)** 4 : 1 **(d)** 1 : 1

13. Water has different properties than hydrogen and oxygen because it is a: 1

(a) mixture **(b)** compound **(c)** alloy **(d)** element

14. The floating of ice is important because: 1

(a) it makes water salty
(b) it removes oxygen
(c) it heats the lake
(d) it insulates water below for aquatic life

15. A concave lens makes light rays: 1

(a) converge **(b)** diverge **(c)** stay parallel **(d)** stop

16. The Surya Siddhanta is an: 1

(a) river **(b)** modern calendar
(c) ancient Indian astronomical text **(d)** satellite

17. A tapeworm in the gut of a human is an example of: 1

(a) parasitism **(b)** commensalism **(c)** predation **(d)** mutualism

18. Organisms that eat only animals are: 1

(a) producers **(b)** herbivores **(c)** decomposers **(d)** carnivores

19. In birds, development of the zygote into an embryo occurs: 1

(a) after the egg is laid **(b)** in the soil only
(c) entirely inside the mother **(d)** in water

20. Which action helps protect the Earth? 1

(a) Reuse, repair and recycle **(b)** Cut forests
(c) Dump waste in rivers **(d)** Burn plastic

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Why are viruses different from bacteria? 2

22. Mention two ways to reduce antibiotic resistance. 2

23. What did Galvani and Volta disagree about? 2

24. What causes lightning? 2

25. What are minerals? 2

26. Why does soda lose its fizz faster when warm? 2

27. Name two types of protected areas in India. 2

28. Define pollination and fertilisation. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. State how Grade 6, 7 and 8 books of Curiosity differ in their message. 3

30. Is gravitational force a contact force? Justify. 3

31. Can a sample of water be a solid, a liquid and a gas? Explain. 3

32. What is a diverging mirror and a diverging lens? Give one example each. 3

33. What is the difference between rotation and revolution of Earth? How do they define day and year? 3

34. Explain biodiversity loss as one part of the triple planetary crisis. 3

35. How does Earth behave like a magnet? 3

OR

Differentiate between a dry cell and a rechargeable battery. 3

36. Why should local communities participate in conserving ecosystems? 3

OR

What is the role of abiotic components in an ecosystem? 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 80–100 words each.

37. Describe Alexander Fleming's discovery. How do antibiotics differ from vaccines? 5

OR

Compare acquired immunity from infection and from vaccination. 5

38. Calculate pressure for 2 situations: a 40 N brick on its 0.1 m² face and on its 0.05 m² face. Explain the difference. 5

OR

Explain local storms of India: Kalboishakhi, Bordoila and mango showers. State their usefulness. 5

39. Describe how a solute behaves when a saturated solution is cooled and show it with a crystallisation example. 5

OR

Write a note on the importance of solutions in nature and industry. 5

40. Describe the work of conservation: national parks, sanctuaries, biosphere reserves and community areas. 5

OR

A pond has algae, small fish, big fish and bacteria. 5
(i) Make a food chain.
(ii) Name the producer and consumers.
(iii) Role of bacteria.
(iv) Effect of removal of small fish.

END OF QUESTION PAPER — Total: 80 Marks